

Dr Mahesh Joshi



CONTACT INFORMATION

Email: mahesh.joshi@iiitp.ac.in, maheshjnvw@gmail.com

Contact number: (+91) 8886844737

Address: A-604, Silver Springs Phase II,
AB Bypass Road, Nayta Mundla,
Indore, Madhya Pradesh, India 452020

RESEARCH INTERESTS

Biometric Security, Internet-of-Things (IoT) Security

RESEARCH PROFILES

- Researcher ID (WoS/Thomson/Publons): **IAO-4226-2023**
- SCOPUS ID: **57218293162**
- ORCID ID: **0000-0002-4814-9555**
- Vidwan ID: **377724**
- Google scholar: **bxvHCsYAAAAJ**
- Research Gate: **Mahesh_Joshi27**
- LinkedIn: **mahesh-joshi-669464223**

EDUCATION

Indian Institute of Technology (IIT), Indore, India

PhD, Computer Science and Engineering,

- Thesis title: *Mitigating MasterPrint Vulnerability A Security Threat To Fingerprint Biometric Systems*
- Advisor (1) : **Dr. Bodhisatwa Mazumdar**
- Advisor (2) : **Dr. Somnath Dey**

Visveswaraya National Institute of Technology (VNIT), Nagpur, India

M.Tech., Computer Science and Engineering,

- Thesis title: *Automatic Parallelization for Multi-core Architecture using OpenMP*
- Advisor : **Prof. S. R. Sathe**

Sant Gadge Baba Amravati University, Amravati, India

B.E., Information Technology,

CERTIFICATIONS

Certified IBM Cyber Security Analyst

WORK EXPERIENCE

Assistant Professor

Indian Institute of Information Technology,
Pune, India

Assistant Professor

Dr Vishwanath Karad MIT World Peace University,
Pune, India

Assistant Professor

Vignan's Foundation for Science Technology and Research University,
Guntur, India

Assistant Professor

St. Vincent Pallotti College of Engg. and Tech. (Autonomous),
Nagpur, India

ADMINISTRATIVE DUTIES **Institute level**

1. **Assistant Director,**
APSSDC-tSDI (Andhra Pradesh State Skill Development Corporation - technical Skill Development Institute), VFSTR University, Guntur.

Department level

1. **Department Magazine Coordinator,**
Dept. of CSE, VFSTR University, Guntur.
2. **GATE Coordinator,**
Dept. of CSE, VFSTR University, Guntur.
3. **Technical Skill Committee Member,**
Dept. of CSE, VFSTR University, Guntur.
4. **Exam Section Faculty In-charge,**
Dept. of CSE, VFSTR University, Guntur.

PUBLICATIONS
(PUBLISHED IN
JOURNALS)

1. **Mahesh Joshi**, Bodhisatwa Mazumdar, and Somnath Dey. *A Novel Minutiae-oriented Approach for Partial Fingerprint-based MasterPrint Mitigation*. Pattern Recognition, Volume 145, January 2024, 109935
<https://doi.org/10.1016/j.patcog.2023.109935>
(ISSN: 1873-5142, Elsevier, SCI indexed, IF = 8)
2. **Mahesh Joshi**, Bodhisatwa Mazumdar, and Somnath Dey. *A comprehensive security analysis of match-in-database fingerprint biometric system*. Pattern Recognition Letters 138, (2020), 247–266.
<https://doi.org/10.1016/j.patrec.2020.07.024>
(ISSN: 0167-8655, Elsevier, SCI indexed, IF = 5.1)
3. **Mahesh Joshi**, Bodhisatwa Mazumdar, and Somnath Dey. *Mitigating MasterPrint vulnerability by employing minutiae geometry*. J. Electron. Imaging 31(1), 013026 (2022),
<https://doi.org/10.1117/1.JEI.31.1.013026>
(ISSN: 1017-9909, SPIE, SCIE indexed, IF = 1.1)
4. **Mahesh Joshi**, Bodhisatwa Mazumdar, and Somnath Dey. *Investigating the impact of thresholding and thinning methods on the performance of partial fingerprint identification systems: a review*. J. Electron. Imaging 32(1), 010901 (2023),
<https://doi.org/10.1117/1.JEI.32.1.010901>
(ISSN: 1017-9909, SPIE, SCIE indexed, IF = 1.1)

BOOK CHAPTERS
PUBLISHED

1. **Mahesh Joshi**, Bodhisatwa Mazumdar, and Somnath Dey. 2021. *Biometric-based Secure Authentication for IoT-enabled Devices and Applications*. In (First) Security of Internet of Things Nodes Challenges, Attacks, and Countermeasures. Chapman, Boca Raton, Florida, 81–106., ISBN: 978-0-367-65049-0
<https://doi.org/10.1201/9781003127598>
2. **Mahesh Joshi**, Bodhisatwa Mazumdar, and Somnath Dey. 2021. *Lightweight Security Protocols for Securing IoT Devices in Smart Cities*. In Data-Driven Mining, Learning and Analytics for Secured Smart Cities: Trends and Advances. Springer International Publishing, Cham, 89–108., ISBN: 978-3-030-72138-1
https://doi.org/10.1007/978-3-030-72139-8_5
3. **Mahesh Joshi**, Bodhisatwa Mazumdar, and Somnath Dey. 2021. *Advance Security Schemes for Smart Societies*. In Green Technological Innovation for Sustainable Smart Societies: Post-Pandemic Era. Springer International Publishing, Cham, 301–322., ISBN: 978-3-030-73294-3
https://doi.org/10.1007/978-3-030-73295-0_14

CONFERENCE
PAPER

1. **Mahesh Joshi** and Bodhisatwa Mazumdar and Somnath Dey, *On the prospects of latent MasterPrints*. 6th IAPR International Conference on Computer Vision and Image Processing. 3 – 5th December 2021, IIT Ropar, Punjab, India. ISBN: 978-3-031-11345-1
https://doi.org/10.1007/978-3-031-11346-8_49

POSTER
PRESENTATION

1. **Mahesh Joshi**, Bodhisatwa Mazumdar, and Somnath Dey. *Biometric Security and Authentication: Threats, Countermeasures and Template Protection Schemes*. 10th International Conference on Security, Privacy and Applied Cryptographic Engineering (SPACE'20), 17 – 20th December 2020, IIT Kharagpur.

COMMUNITY
SERVICES

Journal Editorial Board member

1. Section Editor, *Medical Imaging Process & Technology (MIPT)*, EnPress Publisher, LLC, ISSN: 2578-160X (Online)

Journal reviewer

1. *IEEE Computer Magazine* (Science Citation Index Expanded (SCIE) indexed)
2. *Security and Communication Networks* (Science Citation Index Expanded (SCIE) indexed)
3. *SN Computer Science* (SCOPUS indexed)
4. *International Journal of Electrical and Computer Engineering* (SCOPUS indexed)
5. *Journal of Autonomous Intelligence* (SCOPUS indexed)
6. *Recent Advances in Electrical & Electronic Engineering* (SCOPUS indexed)

Conference reviewer

1. 2nd IEEE International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME'22) (Maldives, 16 – 18th November 2022)
2. 2022 3rd International Symposium on Automation, Information and Computing (ISAIC 2022) (Beijing Jiaotong University, China, 9 – 11th December 2022)

3. IEEE ICECET'22 (Prague, Czech Republic, 20 – 22th July 2022)
4. IEEE ANTS' 2021 (Hyderabad, India, 13 – 16th Dec. 2021)
5. IEEE ICECET 2021 (CapeTown - South Africa, 9 – 10th Dec. 2021)
6. IEEE ICECCME 2021 (Mauritius, 7 – 8th Oct. 2021)

Book and book chapter reviewer

1. *Achieving Full Realization and Mitigating the Challenges of the Internet of Things*, IGI Global.
2. *From Internet of Things to Internet of Intelligence*, Springer Nature.

WORKSHOP AS CONVENER

1. Workshop on **Python Programming**, VFSTR University, Guntur

WORKSHOP AS INSTRUCTOR

1. Workshop on **Android App Development**, for 250 students from MCA and BBA at MITWPU, Pune

WORKSHOPS/ TRAINING PROGRAMS ATTENDED

1. Attended IET Pune LN-sponsored One-Week FDP on **Mathematical Modeling** at Dr Vishwanath Karad MIT WPU Pune.
2. Completed National Academic Immersion Program (NAIP) in Cybersecurity and Digital Forensics organized by, **Indian Institute of Technology (IIT) Jodhpur Technology Innovation and Start-up Center**.
3. Faculty Development Program (FDP) on **Data Structures and Algorithms, Competitive Coding**, MIT-WPU Pune.
4. One Week FDP on **Role of Teachers in Enhancing Academic and Administrative Excellence**, , MIT-WPU Pune.
5. One day workshop on **Assessment Methodology for Question Paper Setting and Project-Based Learning**, MIT-WPU Pune.
6. Two Week National Academic Immersion Program (NAIP) at, **Internet of Things (IoT) Lab, NITTTR-Siemens Center of Excellence**, NITTTR Bhopal, India.
7. One Week Offline FDP on, **Advances in Cyber Security**, MIT World Peace University, Pune, India.
8. Global Initiative of Academic Networks (GIAN) course, **Ubiquitous Computing**, IIT Indore, India.
9. Global Initiative of Academic Networks (GIAN) course, **Data Protection: from principles to practice**, , IIT Indore, India.
10. Online course on **Intro to Python for Data Science**, at **Datacamp**.
11. One Month **Faculty Industrial Training Program**, at Efftronics Systems Pvt. Ltd., Vijayawada, and implemented an Android App for Controlling Lighting Appliances.
12. Two Days Workshop on **Python Programming**, at VFSTR University, Guntur.

	- Three Months Java Programming Course, at KICIT Nagpur. - Workshop on Linux Internals, at VFSTR University, Guntur. - Two Days Quest for the Best, at SVP CET, Nagpur, India. - FDP Spoken Tutorial based Software Training, at VFSTR University, Guntur. - IUCEE Workshop on Application of Graph Theory and Number Theory in Engineering, at VFSTR University, Guntur. - One Week Workshop on Research Methodology, at PIET Nagpur - One Week Workshop on Machine Learning, at VFSTR University, Guntur. - Workshop on Internet of Things Using Raspberry Pi, at Hyderabad. - One Day Training Program on Stress Management Course, at VFSTR University, Guntur. - Workshop on Mobile Application Development, at VFSTR University, Guntur.
PROFESSIONAL MEMBERSHIP	- European Alliance for Innovation (EAI) - International Association for the Engineers (IAENG) - International Academy for Science & Technology Education and Research (IASTER) - International Technology and Engineering Educators Association (ITEEA)
MASTERS (PG) THESIS SUPERVISION	- Ms Preksha Khant, M. Tech. CSE (Data Science & Analytics)), Dr Vishwanath Karad MIT World Peace University, Pune, Thesis title: Multimodal and Multilabel Movie Genre Classification. - Ms K. Lavanya, M.Tech. CSE), VFSTR University, Guntur, Thesis title: Comparison between CE-stream and ACO algorithms applying on stream clustering data analysis. - Ms V. Harshita, M.Tech. CSE), VFSTR University, Guntur, Thesis title: Hybrid link management source routing protocols for MANETs.
COURSES DESIGNED	UG Level: Cyber Threat Modeling: B.Tech. Cybersecurity and Forensics (2022-26), MIT World Peace University, Pune, India
FELLOWSHIPS RECEIVED	- Cleared the district-level entrance examination for admission in Jawahar Navodaya Vidyalaya (JNV) Washim, India, a boarding school fully funded by Govt. of India. <i>Teaching Assistantship</i> during M.Tech course at VNIT Nagpur from the Ministry of Human Resource Development (MHRD) Govt. of India for being a GATE-qualified candidate. <i>Junior Research Fellowship (JRF)</i> from the Ministry of Education (MoE) Govt. of India for the duration of PhD program at IIT Indore. <i>Senior Research Fellowship (SRF)</i> from the Ministry of Education (MoE) Govt. of India for the duration of PhD program at IIT Indore.

ACCOLADES AND RECOGNITION	1. PhD Thesis accepted in PhD Colloquium in 3 rd IEEE International Conference on Computer Vision and Machine Intelligence (IEEE CVMI) during 19-20 October 2024 at Indian Institute of Information Technology (IIIT), Allahabad.
REFERENCES	Dr Bodhisatwa Mazumdar Associate Professor, Department of CSE, Indian Institute of Technology Indore, India E-mail: bodhisatwa@iiti.ac.in Dr Somnath Dey Associate Professor, Department of CSE, Indian Institute of Technology Indore, India E-mail: somnathd@iiti.ac.in Prof. S. R. Sathe Professor, Department of CSE, Visveswarana National Institute of Technology Nagpur, India E-mail: srsathe@yahoo.com Dr Sridhar Adepu Lecturer, School of Computer Science, University of Bristol, Bristol, United Kingdom E-mail: sridhar.adepu@bristol.ac.uk
DECLARATION	I hereby declare that the above-written particulars are true to the best of my knowledge and belief.